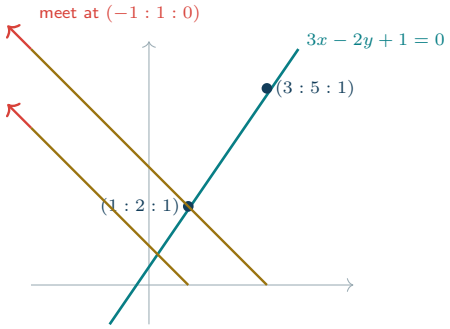
PROJECTIVE GEOMETRY LADDER 2 OF 9

Homogeneous coordinates

A triple up to scale names a point, and one cross product both joins points and meets lines.

1 / THE RULE AND THE PICTURE

Points and lines are both triples up to scale, and incidence is a zero dot product



A point is $(X:Y:Z)$, meaning any non-zero multiple of the triple, and a line is $[a:b:c]$, meaning the equation $aX+bY+cZ=0$, so

join $P,Q:\ P\times Q,$	meet $\ell,m:\ \ell\times m$

The same formula serves both jobs, because points and lines enter symmetrically, and it returns a point with $Z=0$ when the lines are parallel.

2 / THE DICTIONARY

Ordinary points, points at infinity and lines

object	coordinates	example
ordinary point (x,y)	$(x:y:1)$, and $(X:Y:Z)$ with $Z\neq 0$ is $(X/Z,\ Y/Z)$	$(4:6:2)=(2:3:1)$ is $(2,3)$
point at infinity	$(a:b:0)$, the direction (a,b)	$(3:6:0)=(1:2:0)$, slope 2
line $ax+by+c=0$	the coefficients $[a:b:c]$, with $aX+bY+cZ=0$	$x+y=1$ is $[1:1:-1]$
line through two points	the cross product $P\times Q$	$(1:2:1)\times(3:5:1)=[-3:2:-1]$
meeting point of two lines	the cross product $\ell\times m$	$[1:1:-1]\times[1:1:-3]=(-2:2:0)=(-1:1:0)$

3 / WHY THE CROSS PRODUCT WORKS

It is the unique triple orthogonal to both inputs

For points P and Q the vector $P\times Q$ is perpendicular to both, so the line $[a:b:c]=P\times Q$ satisfies $aX+bY+cZ=0$ at P and at Q. The join of $(1,2)$ and $(3,5)$ is $[-3:2:-1]$, which is $3x-2y+1=0$, and indeed $3-4+1=0$ and $9-10+1=0$.
For two intersecting lines the cross product returns an ordinary point, as $[1:1:-1]\times[1:-1:-1]=(-2:0:-2)=(1:0:1)$, the point $(1,0)$ where $x+y=1$ meets $x-y=1$.

4 / COLLINEAR AND CONCURRENT

A determinant detects both

Three points are collinear exactly when the determinant of their coordinate triples is zero, and three lines are concurrent exactly when the determinant of their coefficient triples is zero. This turns incidence statements of geometry into polynomial identities, which is how the theorems of Pappus and Desargues are checked by computer on later sheets.

5 / TRY THESE

Compute with the triples

(a) Find the line through $(0,0)$ and $(2,1)$: $(0:0:1)\times(2:1:1)=(-1,2,0)$, so $[-1:2:0]$, which is $y=x/2$. **(b)** Where do $x=1$ and $x=3$ meet? $[1:0:-1]\times[1:0:-3]=(0,2,0)$, so $(0:1:0)$, the vertical direction. **(c)** Why is $(2:4:6)$ the same point as $(1:2:3)$? They are multiples of one another.

ASIDE / DUALITY

Swap the words point and line in a true statement and it stays true, because the same cross product underlies both.

LINKS TO OTHER SHEETS

The previous sheet explains the point at infinity; the next sheet writes curves in these coordinates.

Convention. $(X:Y:Z)$ is a point; $[a:b:c]$ is a line; $P\times Q$ is the cross product of the coordinate triples; the determinant test for collinearity is $\det(P,Q,R)=0$.